POS1
C – Practice Exercises
Year 1 / Q1

Introduction
Data Types

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1. Introduction

Q1_101_Welcome

Write a C-program **Welcome**. It should print the text "**Welcome 1AHIF**" to the console. Use a **printf(...)** statement.

Q1_102_Output

Create a C-program **ConsoleOutput**. Following text is printed to the console.

```
Name: Maxi Mustermann
Class: 1AHIF
Catalog number: 21
Date: 12.10.2018
```

Q1_103_InputOutput_I

Create C-program **InputOutput_I.** In the **main()** function, the **scanf**() reads a first and last name on two character fields **firstName** and **lastName**. The two names are then output to the console.

```
Please enter your first name : Maxi
Please enter your last name: Mustermann
-------------------------------------------------------------------
The name is: Maxi Mustermann
```

Note: `char text[50] = ;`

Q1_104_InputOutput_II

This program has the same functionality as Q1_103_InputOutput_I: In the **main()** function **gets()** reads a first and last name on two character fields **firstName** and **lastName**. The two names are then output to the console.

Q1_105_InputOutput_III

Create the C program **InputOutput_III**. Declare an **int** variable **n**, a **float** variable **x**, a **double** variable **y**, a **char** variable **c** and a **string str** for 30 characters. Read values for all variables from the console. Then output the variables back to each other. Use the functions **scanf()** and **printf()**.

```
Please enter an int value: 5
Please enter a float value: 2.1
Please enter a double value: 3.14
Please enter a char value: A
Please enter a string: Hello
-----------------------------------------
n = 5
x = 2.1
y = 3.14
c = A
str = Hello
```

2. Formatted output

Q1_201_ OutputFormat

Create a C-program `OutputFormat.` Declare an `int` variable `n`, a `float` variable `x`, a `double` Variable `y`, a `char` variable `c`, and a `string str` for 30 characters.
Assign them the values 5 / 1.234567 / 1.234567 / A / Hello.

• `int` variable `n`:
– Width 8, right-aligned
– Width 8, left-aligned
– Width 8, right-aligned with leading zeros
– Width 8, left-aligned with sign

• `float` variable `x`:
– Width 10, 4 decimal places, right-aligned
– Width 10, 2 decimal places, left-aligned
– Width 10, 3 decimal places, right-aligned with leading
– Width 10, 3 decimal places, right-aligned with leading zeros and signs

• `double` variable `y`:
– Width 10, 4 decimal places, right-aligned
– Width 2, 4 decimal places, right-aligned
– Width 15, 8 decimal places, right-aligned
– Width 10, 2 decimal places, left-aligned
– Width 10, 3 decimal places, right-aligned with leading zeros

• `char` variable `c`:
– Width 3, right-aligned
– Width 3, left-aligned

• `String str`:
– Width 10, right-aligned
– Width 3, right-aligned
– Width 10, left-aligned

Q1_202_Fix_an_error

Correct the errors in the program:

```
void main(){
  int ch, a;
  scanf("Los mit einer Taste und Return: %ch\n", &c);
  printf("\nGeben Sie nun eine ganze Zahl zwischen 1 und 200ein: ");
  scanf("%d",a);
  printf("\nIhre Eingabe: %d", &a);
}
```

Q1_203: What does that mean?

Write down the output of the following program piece on the screen. Specify exactly all spaces! If necessary, check your answer on the computer.

```
int z=125;
char ch = 'X';
printf("Wert=%+06d",z);
printf("%-7.2f\\%7.2f",w,w);
printf("%c-%d-%3c-%5d",z,z,ch,'A');
```

3. Integer data types: byte, short, int, long

Q1_301_UpDownCounter

Create a C-program `UpDownCounter` that reads a number, increments it 3 times by one, and outputs the number to the console.

Example:

```
Please input a number: 21
The present value of the number is: 22
The present value of the number is: 23
The present value of the number is: 24
```

Q1_302_PocketCalculator

Create a C-program `PocketCalculator` that reads two numbers, adds them, subtracts them and outputs the result. Use the attributes `number1`, `number2`, `sum`, `difference`. The format of your output should look like that in the example.

Example:

```
Input the first number  : 21
Input the second number :  5
Addition   : 21 + 5 = 26
Subtraction: 21 - 5 = 16
```

Q1_303_Rectangle

Create a C-program `Rectangle` that reads in the width and height of a rectangle, computes the perimeter and the area, and prints the results to the console. Use the attributes `height, width`, `perimeter` and `area`. The format of your output should be like that in the example:

Example:

```
Width of the rectangle: 10
Height of the rectangle: 6
Perimeter: 32 cm
Area: 60 cm2
```

Q1_304_Cube

Create a C-program `Cube` that reads in the length of the side of a cube. It then computes the surface, the volume of the cube and prints the result to the console. Use the attributes `side, surfaceArea` and `volume`. Format the output as in the following example:

Example:

```
Length of the side of the cube: 5
Surface: 150 cm2
Volume: 125 cm3
```

Q1_305_SecondsCalculator

Create a C-program `SecondsCalculator` that reads in the hours, minutes, and seconds from the console. It then computes how many seconds have passed since midnight and outputs this number:

Example:

```
Input the hours: 1
```

```
Input the minutes: 1
Input the seconds: 1
3661 seconds have passed since midnight
```

Q1_306_TimeComputer

Create a C-program `TimeComputer` that reads the number of seconds that have passed since midnight from the console. It then computes the number of hours, minutes and seconds for this value of time and prints the output in time format as shown in the example.

Example:

```
Input the number of seconds since midnight: 3661
Time: 1:1:1
```

Q1_307_WeeklySalary

Create a C-program `WeeklySalary.` It reads the hourly rate and the hours worked from the console and computes the weekly salary. Assume that your employee always works at least the basic workload of 40 hours per week. Extra hours count double. So, we get the following formula for the weekly salary:
weekly salary = 40 * hourly rate + extra hours * hourly rate * 2
Output the weekly salary you get:

Example:

```
Input the hourly rate in Euros: 20
Input the extra hours worked: 10
Your salary is 1200.- EUR
```

Q1_308_SumOfDigits1

Create a C-program `SumOfDigits1` that read in a two-digit integer number, computes the sum of the two digits and outputs the result. To find the two digits use division "/" and modulo "%".

Example:

```
Input a two-digit number: 67
The sum of the digits is: 13
```

Q1_309_SumOfDigits2

Same as example SumOfDigits1, except do it for three digits.

Example:

```
Input a three-digit number: 234
The sum of the digits is: 9
```

Q1_310_ChangeCalculator

Create a C-program `ChangeCalclulator` that reads in an amount in Euros and computes the change (germ. Wechselgeld). The change should have as few coins as possible. Possible coins are 1€- and 2€-coins and paper money for 5€-, 10€-, 20€-, 50€- and 100€.

Example:

```
Please input the amount in Euros: 47
47 EUR will be changed to:
```

```
0 x 100 EUR
0 x  50 EUR
2 x  20 EUR
0 x  10 EUR
1 x   5 EUR
1 x   2 EUR
0 x   1 EUR
```

Q1_311_OrderOfMagnitude (germ. Stellenwert)

Create a C-program `OrderOfMagnitude` that inputs a 5-digit number and prints the digits according to their positional order of magnitude. Use the attributes `inputNumber`, `onesPlace`, `tensPlace`, `hundredsPlace`, `thoudsandsPlace`, and t, … `stenThousandsPlace`.

Example:

```
Please input a five-digit number: 45231
ten thousands: 4
thousands    : 5
hundreds     : 2
tens         : 3
ones         : 1
```

Q1_312_LottoNumbers

Create a C-program `LottoNumbers.` It computes 7 Lotto numbers in the range 1 bis 45. Use the attributes `lotto1`, … `lotto7` to store the 7 random numbers. Print the values in the format shown in the example:

Example:

```
The following lotto numbers have been drawn: 21, 6, 8, 42, 35, 1
Additional number: 14
```

Q1_313_FractionCalculator

Create a C-program FractionCalculator that reads in the numerator and denominators of two fractional numbers. It then outputs the result of adding and subtracting the two numbers. Print the result in the format shown in the example.

Example:

```
Number 1 - numerator: 1
Number 1 - denominator: 2
Number 2 - numerator: 3
Number 2 - denominator: 4

Addition: 1/2 + 3/4 = 10/8
Multiplication: 1/2 * 3/4 = 3/8
```

4. Floating point data types: `float`, `double`

Q1_401_CircleCalculator

Write a C-program `CircleCalculator` which reads in the radius of a circle and then computes the absolute value of the radius, the perimeter and the area. The output should be formatted to an accuracy of 2 decimal points. All attributes are of type float. Use the functions `pow()` and the constant `M_PI` for your calculations (therefor you have to include the file math.h). To format your output properly use the format specifier of `printf()`.

Example:

```
Enter the radius: 3.0
Perimeter: 18.85
Area: 28.27
```

Q1_402_Triangle

Of a right triangle, the side opposite the right angle is called the hypotenuse. The sides adjacent to the right angle are called *legs* (or *catheti*, singular: cathetus). Write a C-program `Triangle` to read in the two legs of a triangle, to compute the hypotenuse, and the perimeter and the area of the triangle. The results are printed to the console. Use variables of the type `double` and the mathematic functions `pow()` and `sqrt()` for the computation of the square and the square root. The results should be printed to one decimal point.

Example:

```
Length of side a: 3
Length of side b: 4
The length of the hypotenuse is 5.0 cm
The perimeter is 12.0 cm
The area is 6.0 cm2
```

Q1_403_Annulus

Annulus (germ. Kreisring): Write a C-program `Annulus,` that reads in the radius of two concentric circles. Compute the perimeter and the area of the resulting annulus. The perimeter is the sum of the perimeters of the two circles, the area is the difference of the larger and the smaller area. The output should be formatted to an accuracy of three decimal places.

Example:

```
Radius outer circle: 5
Radius inner circle: 4
Annulus perimeter: 56.549 cm
Annulus area: 28.274 cm2
```

Q1_404_InterestCalculator

Write a C-program `InterestCalculator,` that prompts the user for nominal capital, interest, and duration (germ. Grundkapital, Zinssatz, Laufzeit) for computation of the compound capital. You can use the following formula:

capital = nominal * r^n,

where

r = 1 + interest / 100 , n = duration in years

Example:

```
Nominal capital: 1500
Duration in years: 30
Interest in %: 2.5
Compound capital after 30 years: 3146.35 EUR
```

Q1_405_CreditCalculator ***difficult!***

Write a C-program `LoanCalculator` to calculate the final sum of a loan. The class takes as input the monthly installment payment, the term of the loan in months and the monthly interest rate. Use the following formula:

Loan = Payment * $(1-q^n) / (1-q)$

where q = 1 + rate/ (12*100)

n = number of months

Example:

```
Monthly rate in EUR: 500
Term in months: 60
Interest rate in %: 4.5
Total capital paid after 60 months: 33572.77 EUR
```

Q1_406_TempCalculator

Write a C-program `TempCalculator` to convert Celsius to Fahrenheit and vice versa. First, a temperature value is read in as Fahrenheit, converted to Celsius and printed to an accuracy of 2 decimal places. Then a Celsius value is read in and converted to Fahrenheit.

The formula for converting Fahrenheit (F) to Celsius © is : C = 5 / 9 * (F-32)

Example:

```
Temperature in F: 85
85 Degrees Fahrenheit is 29.44 Degrees Celsius
Temperature in C: 30
30 Degrees Celsius is 86 degrees Fahrenheit
```

Q1_407_SphereComputer

Write a C-program `SphereComputer` to calculate the surface area and volume of a sphere. Read in the radius of the sphere. Use the data type double and the functions `pow ()` and `sqrt()`. If you do not know the formulas for calculating the surface area and volume are, search for them on the web.

Example:

```
Input the radius of the sphere: 1
Surface area: 12.56 cm2
Volume: 4.18 cm3

Input the radius of another sphere: 5
Surface area: 314.15 cm2
Volume: 523.59 cm3
```

Q1_408_Pyramide

Write a C-program `Pyramid` to calculate the surface area and volume of a square pyramid. Read in the side length of the base area and the height of the pyramid. The output should be is formatted to 2 decimal places. Search the formulas for the calculation on the web.

Example:

```
Side length of the base: 1
Height of the pyramid: 1
Surface area of the pyramid: 3.24 cm2
```

```
Volume of the pyramide: 0.33 cm3
```

Q1_409_Cylinder

Write a C-program `Cylinder` for the calculation of the area of the side (*germ. Mantelfläche)*, total surface area (*germ. Oberfläche*) and volume of a cylinder. Generate random numbers for the radius and height of the cylinder. These number must be in the range 1.0 ... 5.0 and must be produced exactly to one decimal place. The output is formatted as shown in the example. Search the formulas for the calculation on the web.

Example:

```
Cylinder computation:
--------------------------------
Radius:  3.4
Height:  4.8

Results:
Area of side: 102.54 cm2
Surface area: 175.18 cm2
Volume:       174.32 cm3
--------------------------------
```

Q1_410_FuelComputer

Write a C-program `FuelComputer` that calculates the average fuel consumption of a car. A driver starts his trip with a full tank at an initial mileage (input). After completion of his journey, the driver inputs the new mileage. He wants to know how many liters are missing in the tank. The car consumes 7 liters of gasoline per 100 km.

Example:

```
Input:
--------------------------------
Mileage at start: 30500
Mileage at end  : 31000
--------------------------------
Result:
Km driven: 500
Fuel consumption [in Liter]: 35
--------------------------------
```

Q1_411_Rounding

Write a C-program `Rounding` that generates a random number of type double in the range of 0 to 5. Then round the random number to n decimal places, where n is input by the user. Output of the number to 8 decimal places.

Hint: $zz = zz*10^n$, cut off the decimal point, then $zz = zz / 10^n$

Example:

```
Generated random number: 3.42147361
Input the number of decimal points: 4
Rounded random number: 3.42150000
```

Q1_412_DecimalHour

Write a C-program `DecimalHour` that reads a time as hour (0 – 24), minutes (0-60), and seconds (0-60). Then it calculates the time in decimal. Output both the input data (hour, minute, second) and the decimal result.

Hint:

Decimal minutes = minutes / 60

Decimal seconds = second / 3600

Example:

```
hours: 6
minutes: 30
```

```
seconds: 30
The time 06:30:30 equals 6.50833 decimal hours
```

Q1_413_ AverageCalculator

Write a C-program `AverageCalculator,` which reads 5 decimal numbers on variables of the data type double, calculates and outputs the sum of these numbers and the arithmetic mean. The mean is calculated from the sum divided by the number of input numbers:
Example:

```
Input:
======================
Number1: 12.3
Number2: -298.349
Number3: 424.24
Number4: 7.5691
Number5: -129.6
======================
Sum:     16.1601
Mean:     3.2320
```

Q1_414_ReverseNumbers

Write a C-program `ReverseNumbers,` that takes the four numbers as input and outputs them in reverse order. Hint: you only need to swap `number4` and `number1`, followed by swapping `number2` and `number3`!
Example:

```
>>Input>>
Number 1: 3
Number 2: 8
Number 3: 15
Number 4: 24
>>The original numbers         >> 3, 8, 15, 24
>>The numbers in reverse order >> 24, 15, 8, 3
```

Q1_415_FunctionComputer

Write a C-program `FunctionComputer` which computes a table of values for a quadratic function of the form y = y = a*x^2 + b*x + c. For the function parameters a, b and c random numbers from 0 to 10 are generated. The program reads the start value for the variable x, computes the resulting function value y. This calculation is repeated four times, with the value of x increasing each time by 1. The output should be as shown in the example.

Example:

```
Enter the starting value for x: 3.2
========================================
Function parameters:
a = 4
b = 6
c = 2

Table of values:
   x  |   y
------+-------
 3.20 | 62.16
 4.20 | 97.76
 5.20 | 141.36
 6.20 | 192.96
 7.20 | 252.56
```

5. Character type: char

Q1_501_Encoder

Write a C-program `Encoder` that reads a string from the console into an array `inString` of type `char` and encodes it into an integer. The first three characters should be stored to variables of type `char`. Then the ASCII values of these three characters should be concatenated to an integer number. For instance, if `inString` is “ABCDEF”, the first three characters are 'A', 'B', and 'C'. Their ASCII values are 65, 66, and 67. This results in the integer number `656667`.

Example:

```
Enter your string: ABC
The encoding is: 656667
```

Q1_502_Decoder

Write a C-program `Decoder`, reversing the procedure of the previous example. Read a 6-digit number from the console. Then group two consecutive digits resulting in three pairs of digits. Then convert each pair into the corresponding ASCII code and store the ASCII codes to three variables of type `char`. After decoding the 3 `char`'s are concatenated into one string using an array of type `char`. Print this array to the console.

Hint: you can use the following code to store a `char` Variable to an array at index 0.

```
char c = 'x';
char text[5] = "";
text[0] = c;
```

Example:

```
Enter your code: 656667
The decoded string: ABC
```

Q1_503_Encryption

Write a C-program `Encryption`, reading a string with 5 Letters (uppercase and lowercase) from the console. For each character in this string, you take the subsequent ASCII character from the ASCII table to replace the original character. Store the replacement characters to a **new** string.

Example:

```
Input string: Hallo
Encrypted string: Ibmmp
```

Q1_504_Acronym

Write the class `Acronym` that reads 3 words (maximum 10 characters per word) from the console and stores them in 3 different arrays of type `char` (choose array size 11). The initial characters of these words are then concatenated to a new string `acronym` and printed to the console.

Example:

```
Input text (enter 3 words):
Check
all
things

Acronym: Cat
```

Q1_505_ CharGames

Create the C program `CharGames`, which reads a string of length 6 from the console.
Create a second string that stores the letters of the first string in reverse order.
Then swap the 1st and 3rd characters as well as the 2nd and fourth characters in the first string. Print the modified string.

Example:

```
Your string: Flower
Second string: rewolF
Your modified string: owFler
```

Q1_506_ CharAndDigit

Create the C program `CharAndDigit`, which reads a two-digit number from the console. Determine the ASCII sum of the two digits of this number and output it.
Now create two int random values in the range of the ASCII values of the characters '0' and '9'. Pictures from the two random values the corresponding two-digit numerical value and output it.

Example:

```
Your number: 37
ASCII value of number 3: 51
ASCII value of number 7: 55
The ASCII sum of the two digits is: 51 + 55 = 106

First random number: 50
Second random number: 49
The character for the ASCII value 50 is '2'.
The character for the ASCII value 49 is '1'.
New number: 21
```

Q1_507_ UpperToLower

Create the C program `UpperToLower`, which reads a string consisting of 4 uppercase letters. Replace all uppercase letters by the corresponding lowercase letters.
Now read in a second string, which consists of 4 lowercase letters. Replace all lowercase letters with the corresponding capital letters.

Example:

```
First string: HOUSE
Changed string: house
Second string: tree
Changed string: TREE
```

Q1_508_RandomWord

Create the C program `RandomWord`, which generates a random word consisting of 3 characters. The random word should consist of a random uppercase letter, a random lowercase letter and a random number.

Example:

```
Random word: Bt7
```

Addition:
Read the uppercase letters, lowercase letters, and numbers that serve as barriers to random value determination.
Example:

```
Lower barrier for the capital letter: X
```

```
Upper barrier for the capital letter: Z
Lowercase barrier: a
Upper limit for the lowercase letter: d
Base barrier for the number: 3
Upper barrier for the number: 6
Random word: Ya5
```